

Midhun Rajan

Manufacturing & Process Engineer | Manufacturing Systems | Digital Transformation & AI Automation

Luton, Bedfordshire, UK | midhunrajan74@gmail.com | +44 7445 588889

Portfolio: mr4693.github.io/Midhun-Portfolio | github.com/MR4693 | linkedin.com/in/midhun-rajan

PROFESSIONAL SUMMARY

Manufacturing and process engineer with experience across sheet metal design, fabrication and ERP-driven production in India, the UAE and the UK since 2017. Today I work inside a live precision sheet metal manufacturing environment, with a dual focus on production engineering and the software systems that run the floor. Over 2025 I led a digital transformation programme that moved core operations from paper to a substantially digitised, ERP-integrated production environment — specifying, designing, validating and deploying four systems, built using AI-assisted software development under my own engineering direction, now in active daily use across 19 operators and 14 work centres.

The work spans the full path from operational problem to deployed system: shop floor digitisation, real-time production intelligence, and the automation of engineering document and job pack workflows — all integrated against MIE Trak ERP on factory infrastructure. Every requirement was identified through direct operational involvement, and I continue to maintain and improve each system in daily production use.

EXPERIENCE

Manufacturing & Process Engineer	2025 - Present
Technical Engineer	Nov 2022 - 2025

Ludwick Precision Ltd · Precision Sheet Metal Manufacturing · Hitchin, Hertfordshire, UK

- ✓ Manufacturing engineering across a multi-work-centre shop floor — CNC fibre laser cutting, tube laser processing, CNC press brake, welding, fabrication and assembly — including sheet metal modelling, DFM review and ERP-driven production planning.
- ✓ Led an end-to-end digital transformation programme, identifying operational problems from daily floor involvement and delivering four ERP-integrated systems rather than purchasing commercial software — using AI-assisted software development under my own engineering direction.
- ✓ Designed and deployed the **Manufacturing Operations Platform** — Android shop floor terminals for clock-on, digital job packs, live work orders, move tickets and sign-off — eliminating paper documentation and end-of-shift ERP data entry across 19 operators.
- ✓ Designed and delivered the **Manufacturing Intelligence Platform** — real-time management dashboards over live MIE Trak data — giving directors and supervisors continuous shop floor visibility without ERP access or floor walks.
- ✓ Designed and delivered the **Digital Job Pack Platform** — Excel-driven ERP lookup with automated PDF extraction and assembly — reducing job pack preparation from hours to seconds per batch.
- ✓ Designed and delivered the **Document Intelligence Platform** — drag-and-drop intake with automatic part number detection and automated MIE Trak ERP filing for PDF, STEP and DXF files — replacing manual document control with a system-enforced process.
- ✓ Defined requirements, designed the workflows and business logic, validated and tested every feature, and deployed each system into live operations — and continue to maintain and improve all four on factory infrastructure, with no external development resource.

Design Engineer

Apr 2021 – Jun 2022

Dolphin Manufacturing LLC · Sheet Metal Manufacturing · Ajman, UAE

- ✓ Designed sheet metal components and assemblies for manufacture using 3D CAD, optimised for laser cutting, bending and fabrication.
- ✓ Produced manufacturing drawings, DXF files and supporting documentation for the shop floor.
- ✓ Worked closely with production teams to ensure designs were practical, manufacturable and cost-effective.
- ✓ Supported product development, design improvements and engineering change implementation.

Fabrication Engineer

Jul 2017 – Aug 2020

R.S. Franklin Associates · Structural & Sheet Metal Fabrication · Nagercoil, Tamil Nadu, India

- ✓ Prepared fabrication and manufacturing drawings for structural and sheet metal projects.
- ✓ Planned fabrication processes and coordinated production activities to ensure quality and on-time delivery.
- ✓ Supported shop-floor operations, material planning and quality inspections.
- ✓ Worked with production teams to resolve manufacturing issues and improve fabrication efficiency.

SELECTED ACHIEVEMENTS

Paperless shop floor — 19 operators and 14 work centres moved onto a live digital platform; paper job packs, clock cards and move tickets replaced with a traceable, ERP-linked workflow.

Real-time ERP data — production records now update the moment operators act, eliminating end-of-shift manual data entry and the data lag that came with it.

Hours to seconds — job pack preparation automated from a 30-minute-to-several-hour manual task to seconds per batch, against live ERP document records.

Management visibility — four-view intelligence platform delivering continuous, real-time production visibility to directors and supervisors with no specialist ERP knowledge required.

EDUCATION

Bachelor of Engineering (Honours), Mechanical Engineering

Anna University · Chennai, Tamil Nadu, India

Master Diploma in Product Design

CADD Centre · Trivandrum, Kerala, India

Professional Diploma in HVAC & MEP Design

CADD Centre · Trivandrum, Kerala, India

SKILLS & CAPABILITIES

Positioned by what I deliver. My core expertise is manufacturing engineering and digital process design; the software was built using AI-assisted development under my own direction. Technical Exposure lists technologies I have practical working knowledge of — not specialist software-engineering expertise.

MANUFACTURING SYSTEMS · CORE EXPERTISE

Manufacturing Process Engineering

ERP Integration (MIE Trak)

SQL Data Integration

Manufacturing Workflow Design

Digital Process Automation

Manufacturing Intelligence

Technical Solution Architecture

AI & Digital Technologies

AI-assisted Software Development

Process Automation

Dashboard Development

Manufacturing Analytics

Document Intelligence

System Integration

Technical Exposure • Working Knowledge

Python

SQL

Flask

HTML

CSS

JavaScript

Manufacturing Disciplines

CNC Fibre Laser Cutting

Tube Laser Processing

CNC Press Brake

Welding & Fabrication

Assembly

SolidWorks

Sheet Metal DFM

Laser Nesting

Production Planning

ERP-Driven Manufacturing

Engineering Approach

On AI-Assisted Engineering

The systems in this portfolio were implemented using AI-assisted engineering, with an AI model used as a software development partner under full engineering direction and accountability. Manufacturing requirements, operational workflows, ERP integration, testing and validation were led by me. AI accelerated software development; the engineering judgement, and accountability for every deployed system, remained mine.